

MnI Theory assignment 4

Calculate the integral $\int f_i d\lambda$ for $i \in \{1,2,3\}$

For the functions

$$\begin{aligned}f_1(x) &= x^2 \\f_2(x) &= \frac{1}{1+x^2} \\f_3(x) &= e^{-|x|}\end{aligned}$$

Before we begin we note that $f_i \in \mathcal{M}(B(\mathbb{R}))^+$ since f_i are continuous in $\mathbb{R}$ and non negative in $\mathbb{R}$

By section 5.7

$$\int f_i d\lambda = \lim_{n \rightarrow \infty} R \int_{-n}^n f_i(x) dx$$

Thus we simply need to calculate some Riemann integrals

1)

$$\begin{aligned}R \int_{-n}^n x^2 dx &= \left[\frac{x^3}{3} \right]_{-n}^n = \frac{n^3}{3} - \left(-\frac{n^3}{3} \right) = \frac{2n^3}{3} \\ \frac{2n^3}{3} &\rightarrow \infty \text{ for } n \rightarrow \infty\end{aligned}$$

2)

We remember that

$$\int \frac{1}{1+x^2} dx = \arctan(x)$$

From analysis 2

Thus

$$R \int_{-n}^n \frac{1}{1+x^2} dx = [\arctan(x)]_{-n}^n = \arctan(n) - \arctan(-n)$$

We note that $\arctan(n) \rightarrow \frac{\pi}{2}$ for $n \rightarrow \infty$, $\arctan(-n) \rightarrow -\frac{\pi}{2}$ for $n \rightarrow \infty$

Thus $\arctan(n) - \arctan(-n) \rightarrow \pi$ for $n \rightarrow \infty$

Thus

$$\lim_{n \rightarrow \infty} R \int_{-n}^n \frac{1}{1+x^2} dx = \pi$$

Also we could remember our cauchy distribution and see that

$$\lim_{n \rightarrow \infty} R \int_{-n}^n \frac{1}{1+x^2} dx = \pi \lim_{n \rightarrow \infty} R \int_{-n}^n \frac{1}{\pi(1+x^2)} dx = \pi$$

Since its a probability distribution

3)

We note that the function is even ie

$$f(x) = f(-x) \forall x \in \mathbb{R}$$

Thus by properties of the Riemann integral

$$\begin{aligned} R \int_{-n}^n e^{-|x|} dx &= 2 \cdot R \int_0^n e^{-|x|} dx = 2 \cdot R \int_0^n e^{-x} dx \\ &= 2 \cdot [-e^{-x}]_0^n = 2 \cdot (1 - e^{-n}) \\ \lim_{n \rightarrow \infty} R \int_{-n}^n e^{-|x|} dx &= \lim_{n \rightarrow \infty} 2 \cdot (1 - e^{-n}) = 2 \end{aligned}$$